RIS / Reconfigurable Metasurface

Weekly Study Plan: RF Foundations + Balanis Antennas + Papers

For a 3rd-year ELEL student at CityU. This revised version removes the heavy Orfanidis-first route and keeps only the RF/microwave part, the Balanis antenna part, and the paper-reading path.

Status note: Orfanidis, Electromagnetic Waves and Antennas, Chapter 1 has already been completed. Since it felt too complicated, this plan does not require continuing Orfanidis for now. Use it later only as a reference, not as the main textbook.

How to use this plan

• Work week by week. Do not try to master every derivation before reading papers.

• For each chapter, aim for usable understanding: definitions, physical meaning, and how the equations are used in RIS/metasurface papers.

• For each paper, write a one-page summary: problem, method, hardware/simulation, main equations, and what you did not understand.

• If a chapter becomes too mathematical, read examples, figures, summaries, and then return later.

Main sources kept in the plan

Layer	Main sources	Purpose
RF / microwave	Pozar, Microwave Engineering	Transmission lines, S-parameters, Smith chart, matching, microstrip, resonators, RF devices.
Antennas	Balanis, Antenna Theory: Analysis and Design	Antenna parameters, array factor, beam steering, microstrip antennas, reflectors.
Papers	Your RIS / metasurface reference list	Connect RF + antennas to real RIS, tunable metasurface, and IR-controlled programmable surface literature.

12-week weekly plan

Week 1 - RF survival toolkit and transmission-line basics

Read	Pozar Ch. 1: Electromagnetic Theory review (skim only what you need). Pozar Ch. 2: Transmission Line Theory.
Focus	Voltage/current waves, characteristic impedance Z0, reflection coefficient Gamma, standing waves, VSWR, input impedance, matched load, short/open load.
Skip / skim	Long field derivations and advanced waveguide material. Your target is not full EM theory yet.
Weekly task	Do 5 simple calculations: Gamma for ZL = 0, infinity, Z0, 2Z0, and Z0/2. Explain each result in words.
Outcome	When a paper shows S11 or reflection phase, you understand that it is describing how much wave comes back and with what phase.

Week 2 - Smith chart, impedance matching, and practical RF intuition

Read	Pozar Ch. 2 sections on Smith chart / input impedance if present in your edition. Pozar Ch. 5: Impedance Matching and Tuning.
Focus	Why matching is needed, return loss, load transformation, quarter-wave transformer, single-stub matching, bandwidth of matching.
Skip / skim	Do not spend too long on every matching network. RIS reading mainly needs the concept, not expert Smith chart design.
Weekly task	Pick three loads and sketch how matching changes reflected power. Write what happens to reflected wave when matching improves.
Outcome	You understand why antennas/metasurface unit cells need impedance control and why reflection can be engineered.

Week 3 - S-parameters and microwave network analysis

Read	Pozar Ch. 4: Microwave Network Analysis.
Focus	S11, S21, S12, S22, scattering matrix, return loss, insertion loss, reciprocal networks, lossless networks, reference impedance.
Skip / skim	Heavy multiport algebra beyond what you need for reading plots.
Weekly task	Take any metasurface/unit-cell paper and identify where it reports reflection magnitude, reflection phase, or S11. Translate the plot into words.
Outcome	You can read simulation/measurement plots from CST/HFSS/VNA-style RF papers.

Week 4 - Microstrip, waveguides, and resonators

Read	Pozar Ch. 3: Transmission Lines and Waveguides, focus on microstrip sections. Pozar Ch. 6: Microwave Resonators, selected parts only.
Focus	Microstrip line, effective dielectric constant, guided wavelength, substrate effects, resonance, quality factor Q, bandwidth.
Skip / skim	Full waveguide modal derivations unless your paper uses rectangular waveguides directly.
Weekly task	For one substrate/frequency, estimate guided wavelength. Then explain why a printed patch or split-ring-like cell can be resonant.
Outcome	You understand why RIS unit cells are often printed metallic patterns on dielectric substrates and why their size is related to wavelength.

Week 5 - RF tuning hardware: diodes, switches, phase control

Read	Pozar Ch. 11: Active RF and Microwave Devices, selected sections on diode/device behavior. Read one PIN diode application note and one varactor diode application note.
Focus	PIN diode as RF switch/current-controlled resistor, varactor as voltage-controlled capacitance, bias network, RF choke, DC blocking capacitor, switchable impedance, phase states.
Skip / skim	Amplifier design, mixer design, oscillator design unless needed later.
Weekly task	Draw two paths for a tunable unit cell: RF path and DC/control path. Label which components block RF and which block DC.
Outcome	You can understand how a theoretical phase state exp(j phi) becomes a real hardware state using diodes, bias, and control circuits.

Week 6 - Balanis antenna basics

Read	Balanis Ch. 2: Fundamental Parameters and Figures of Merit of Antennas.
Focus	Radiation pattern, gain, directivity, efficiency, beamwidth, sidelobes, polarization, radiation intensity, effective aperture.
Skip / skim	Detailed radiation integrals for now.
Weekly task	Choose one antenna datasheet or paper plot. Identify main lobe, sidelobes, beamwidth, gain/directivity, and polarization if given.
Outcome	You can understand antenna performance language used in reflectarray/RIS papers.

Week 7 - Balanis arrays and beam steering

Read	Balanis Ch. 6: Arrays: Linear, Planar, and Circular.
Focus	Array factor, phase progression, element spacing, grating lobes, beam steering, planar arrays, amplitude tapering.
Skip / skim	Advanced synthesis methods at first reading.
Weekly task	Use MATLAB/Python to plot a simple 1D array factor. Change phase shift and show the beam moves. Then try 1-bit phases: 0 and 180 degrees.
Outcome	RIS beam steering becomes physically intuitive: many cells with controlled phases shape the reflected wavefront.

Week 8 - Balanis microstrip antennas and reflector background

Read	Balanis Ch. 14: Microstrip Antennas. Balanis Ch. 15: Reflector Antennas, selected sections.
Focus	Patch antenna resonance, substrate permittivity, fringing fields, bandwidth, feeding methods, aperture idea, reflector/reflectarray intuition.
Skip / skim	Exact closed-form design formulas if they slow you down; understand the physical meaning first.
Weekly task	Estimate rectangular patch dimensions for one frequency and substrate. Then explain how changing geometry or capacitance can shift the phase response.
Outcome	You understand the printed-antenna ancestor of many metasurface/RIS unit cells.

Week 9 - First RIS papers: communication overview

Read	Papers: [25] ElMossallamy et al.; [4] Wu and Zhang; [2] Basar et al.; then [3] Di Renzo et al. selectively.
Focus	What RIS is, why it is useful, passive beamforming, cascaded channel, reflection coefficient per element, channel estimation, hardware constraints.
Skip / skim	Deep optimization proofs and long literature tables at first reading.
Weekly task	For each paper, write: 1) main problem, 2) system model, 3) what each RIS element controls, 4) assumptions, 5) limitations.
Outcome	You can connect RF/antenna ideas to the wireless-communications notation used in RIS papers.

Week 10 - Tunable reflectarray / metasurface papers

Read	Papers: [20] Hum et al.; [21] Sievenpiper et al.; [22] Tian et al.; [28] Kamoda et al.; [29] Yang et al.
Focus	Electronically tunable reflectarrays, tunable impedance surface, 1-bit phase states, reflection phase curves, beam scanning, quantization.
Skip / skim	Manufacturing details that are not central to your project unless your supervisor needs them.
Weekly task	Make a comparison table: paper, frequency, unit-cell type, tuning component, phase states, control method, measured/simulated result.
Outcome	You understand the hardware/antenna-engineering side behind RIS.

Week 11 - IR/light-controlled programmable metasurface papers

Read	Papers: [30] Sun et al.; [10] Sayanskiy et al.; [32] IR communication system; [23] and [24] light-controllable metasurfaces; [31] light-programmable metasurface.
Focus	Remote control, infrared coding, digital coding metasurface, switching states, control architecture, separation of RF surface and control electronics.
Skip / skim	Optical plasmonic details unless your surface operates optically.
Weekly task	Draw a block diagram: controller -> IR transmitter -> receiver/control circuit -> switches/diodes -> RF metasurface state.
Outcome	This week directly supports your FPGA/IR active control research direction.

Week 12 - mmWave, surveys, and final integration

Read	Papers: [1] Rappaport et al.; [6] RIS below 10 GHz survey; [33] intelligent metasurfaces; optional [26] NTN RIS survey if relevant.
Focus	mmWave motivation, blockage, high-gain beams, large arrays, below-10-GHz RIS, intelligent metasurface architecture, research challenges.
Skip / skim	Survey sections unrelated to your frequency band or hardware method.
Weekly task	Write a 2-3 page final summary: What is RIS? How does an RF unit cell control phase? How is RIS different from an antenna array? What hardware limitations matter? Which papers are most relevant to your project?
Outcome	You finish with a coherent map of RF, antenna, and paper knowledge instead of scattered reading.

Paper reading order

Stage	Papers	Read for
RIS overview	[25], [4], [2], [3]	Big picture, system model, passive beamforming, cascaded channel, limitations.
Tunable RF surfaces	[20], [21], [22], [28], [29]	Reflection phase, phase states, unit-cell tuning, beam scanning.
IR/light control	[30], [10], [32], [23], [24], [31]	Control architecture, remote programming, optical/IR switching.
mmWave / surveys	[1], [6], [33], optional [26]	Use-cases, frequency bands, practical challenges, research directions.
Metasurface background	Optional: [11], [16]	Use as reference when papers mention surface impedance, subwavelength unit cells, anomalous reflection.

Chapter checklist

Source	Essential chapters / sections	Why
Pozar	Ch. 1 skim; Ch. 2; Ch. 4; Ch. 5; Ch. 3 microstrip sections; Ch. 6 selected; Ch. 11 selected.	RF language: impedance, Gamma, Smith chart, S-parameters, microstrip, resonance, tuning devices.
Balanis	Ch. 2; Ch. 6; Ch. 14; Ch. 15 selected.	Antenna language: gain, patterns, arrays, beam steering, microstrip antennas, reflector intuition.
Papers	Start with overview papers, then tunable surface papers, then IR/control papers.	Do not start from the most hardware-specific paper before learning RF plots and antenna beam steering.

What counts as completed each week

• You can explain the main idea aloud in 2-3 minutes.

• You can identify the key plots/equations and say what they mean physically.

• You have written one page of notes or completed the weekly task.

• You have a list of unclear points to ask your supervisor, professor, or ChatGPT later.

Do not worry if

• You cannot derive every formula from first principles.

• You only understand 50-60% of a paper on the first pass.

• Balanis/Pozar feel slow. They are reference-style engineering books, not easy tutorial books.

• Your first goal is to build recognition: when you see S11, array factor, reflection phase, 1-bit state, or beam steering, you know what role it plays.